TREE IMPROVEMENT

I. The book on applied tree improvement was written by
Ans : Bruce Zobel and John Talbert

2. Variation within the clonal progenies is due to
Ans Environment

3 Mating designs are termed as incomplete mating design
Ans Polymix

4. More genetic gain can be achieved in the seed orchard
Ans : Clonal

5. Pureline theory was proposed by
Ans : Johannsen

6. Homoclime approach suited to
Ans: Exotics

7. Self incompatibility hampers the development of
Ans : Inbreds

both parent identities is known as
Ans : Fulisib

9, Selection can be resorted for
Ans : Heritable trait

10. Provenance-hybrid seed orchard is used for
Ans : Hybridising diverse and compatible provenance

11. The presence of male and female flowers in different plants is called
Ans : Dioceious

12. Emasculation by suction method is more suitable for
Ans : Small flower

13. In trees, normally the width of pollen dilution zone is
Ans : 400-500 m

14. IBPGR refers to
Ans : InternationalBureau of Plant Genetic Resources

15 proposed the concept of hormonal control of organ formation.
Ans : Skoog and Miller

16. The virus free plant was first produced by
Ans :Cocking and White

17 is a major carbon source in plant tissue culture medium
Ans: Glucose

18 is a method of sterilizing with water vapour under high pressure.
Ans : Autoclaving

19. Cybrid is a
Ans : Somatic embryo

20 mediated transformation is an important indirect gene transfer technique
Ans Agrobacterium

21. The two poly nucleotide strands of the DNA double helix are
Ans : Parallel and Complementary

22. The unorganized mass of tissues are called as
Ans : Callus

23. The original tree (Donor tree) from which parts to be propagated are taken are called
Ans : Ortet

24. Orthotropic growth refers to
Ans : Tree

25. in tree improvement programme, many cycles of selection and breeding refers to
Ans : Recurrent selection

26. The trees produced from the seed of a parent is called
Ans : Progeny

27. Area where superior phenotypes or genotypes are established and managed intensively and entirely for seed production is referred as
Ans : Seed Orchard

28. A condition in which F1 plants of an interspecific cross are vigorous and fertile and their F1 progeny is weak and sterile is called as
Ans : Hybrid breakdown

29. An individual having more than two basic sets of chromosomes is called as
Ans : Polyploids

30, Ability of a gene to affect more than one phenotypic characteristic is
Ans : Pleiotrophism

31. In cross pollinated species a true breeding line obtained by continuous
Ans Inbreeding

32. A plant with functional pollen fails to set seed when self pollinated due to
Ans : Incompatibility

33. Regeneration of whole plant from tissues of an actively dividing plant part such as stem tip, is referred
Ans : Meristem culture

34. Variation that is observed among the plants which are regenerated from callus cultures of somatic explants such as meristem is called
Ans Somoclonal variation

35. Regeneration of plants from isolated meristematic cell or tissues is called
Ans Micro-propagation

36. The plant part used in regeneration in tissue culture is termed as
Ans Explant

37. Isolation, introduction and expression of foreign DNA in plants and animals is
Ans : Genetic Engineering

38. The unique functional position of organisms is
Ans : Niche

39. Speciation occurs through physical separation of population is called
Ans Allopatry

40. Trees that have proven to be genetically superior by means of progeny testing is
Ans : Elite tree

41. Most frequently used seed orchard design is
Ans : Randomized block design

42. Plus tree selection method used in an even aged stand is
Ans : Comparison tree selection

43. The original geographic area from which seed or other propagules were obtained is known as
Ans Provenance

44. Usually, seed production areas should contain a minimum area (ha) of
Ans : 4 ha

45. Notification of seed control order by Government of India in the year
Ans : 1983

46. The seed Act was enacted in the year
Ans 1966

47. The most common plant growth regulators compound used in rooting of tree species is
Ans: IBA

48. For a wind pollinated species, the width of the pollen dilution zone is
Ans : 200-400 m

49. Clonal seed orchard has
Ans : Broader genetic base and less selection differential

50. A tree that has been selected for grading because of the desirable character is _____
Ans Candidate tree

51. Systematic removal of individuals in orchards is called_____
Ans : Rouging

52. Male sterility is used to develop
Ans: Hybrids

53. Seed lot certificate colour is
Ans : Orange or green

54. Difference between seed and grain is
Ans Genetic purity

55. Seed sample certificate colour is
Ans : Blue

56. New genes are created by
Ans : Mutation

57. Possible reasons for seed dormancy is
Ans : Immature embryo

58. Seed drying is very important to maintain its
Ans : Viability and vigour

59. The two most important factors influencing the life span of seeds under storage are

Ans : Relative humidity and temperature

60. Occurance of more than one embryo in seed is known as
Ans : Polyembryony

61. Interaction between alleles of two different loci is called as
Ans : Epistasis

62. Development of embryo without fertilization is
Ans Apomixes

63. The group of related individuals when only one parent is common is called

Ans : Half-sib family

64. Functional unit of inheritance is
Ans Gene

65. Migration of alleles from one population or species into another is called as

Ans Gene flow

66. The performance tree's over a full rotation in the new environment is referred as
Ans : Adaptation

67. The situation in which a vegetative propagule does not assume tree form but continues to grow like a branch is calledas
Ans : Plagiotropic

68. Mating of individuals more closely related by ancestry is called
Ans : Inbreeding

69. When on F1 is crossed with the recessive homozygous parent the cross is
Ans : Test cross

70. The progeny of nucleus seed is called
Ans Breeder seed

71. The individuals of a clone are called
Ans : Ramet

72. Male sterility promotes
Ans : Out breeding

73, Wide crossing is also known as
Ans Distant hybridization.

74. Character showing continuous variation and considerable effect of the environment is
Ans Qualitative character

75. Lack of seed set on self pollination is
Ans : Self incompatibility

76. If all variation was due to genetics, the broad sense heritability would be equal to
Ans : 1
77. Emasculation means
Ans : Removal of male flowers /anthers

78. Genetic gain in tree improvement programme is estimated by the formula
Ans : G = S x h2

79. Which studies are performed in order to describe genetic variation within and between population?
Ans : Population variation

80. Where from the best quality seeds would be obtained for producing quality plating stock?
Ans : Clonal seed orchard

81 Selection in evenaged forest is based on
Ans : Check tree method

82. A cross involving two pairs of gene each pair affecting a different character is called
Ans : Dihybrid

83. More than two alternate form of a gene is
Ans : Multiple alleles

84. The nitrogenous base that is present in RNA but not in DNA is
Ans : Uracil

85. Meiosis takes place in the
Ans : Reproductive cells

86 is used to induce polyploidy in plants
Ans : Coichicine

87. Pangenesis theory was proposed by
Ans : Charles Darwin

88. Nullisomy can be represented by
Ans : 2n - 2

89. The adaptation of whole genotype or gene complex to specific environments is the key characteristic of
Ans : Ecotype

90 exhibits progressive variation
Ans : Cline

91. The method of grading for the uneven aged or mixed species stands is called
Ans : Regression

92. Genetic superiority of an individual is tested through
Ans : Progeny testing

93. Name the authors of the book “Applied Tree Improvement”
Ans : Bruce Zobel and John Talbert

94. The pioneering attempt in tree breeding was made in
Ans : Japan

95. The most important statistical parameter for a breeder is
Ans : Variance

96. Polyploids are common in
Ans Hardwoods

97, The best tool for a tee breeder to increase the genetic gain is
Ans : Selection

98. When germination of all seeds occurs over a relatively brief period it is called
Ans : Quasisimultaneous

99. Physical dormancy is found in
Ans *Acacia nilotica*

100. Water soaking seed treatment is best suited for
Ans *: Grei'illea robusta*

101. Vertical antenna like branches often occurring in conifers outside the normal branch whorl is called as
Ans Ramicorn

102. For a wind pollinated species, the width of the pollen dilution zone is
Ans : 200-400 m

103. Clonal seed orchard has
Ans : Broader genetic base and less selection differential

104. Pollen contamination in seed orchard can be reduced by
Ans : Geographic isolation and supplemental mass pollination

105. Example for closed panmictic population is
Ans : Seed orchard

106. Minimum number of clones to be planted in the first generation seed orchard is
Ans: 25-40

107. Tumbling is the method for
Ans : Extraction of cones

108. Dry strobile is seen in
Ans : Fir

109. Factorial and diallel designs are commonly used in the seed orchards of
Ans : Conifers

110. The species with special orchard problems are
Ans : Dioecious species

111. In prehistoric periods has not been understood or ignored by the foresters
Ans : Parentage

112 variation is due to the interaction of specific alleles at a gene locus
Ans : Dominance

113. Out crossing breeding system maintains a high degree of
Ans : Genetic variation

114 is an example of polyploidy in trees
Ans : Poplars, Giant Sequouia, Red Wood, etc

115. The adaptation of whole genotype or gene complex to specific environments is the key characteristic of
Ans : Ecotype

116 of an individual is defined as twice its GCA
Ans : Breeding value

117. Fungi and bacteria usually enters through the
Ans Stomata

118. Embryo exised test used for surface disinfestations is
Ans: NaOCI

119. Fanning mill is
Ans : Air screen cleaner

120. Monocarpic flower is
Ans : Agave

121. A group of similar plants in structural features and performance is known as
Ans : Variety

122. The seed which farms generally used are called as
Ans : Certified seed

123. Middle layer of fruit wall is called as
Ans : Mesocarp

124. The rudimentary root of the seed or seedling that forms the primary root of the young plant
Ans : Radicle

125. New gene are created by
Ans : Mutation

126. Seed drying is very important to maintain its
Ans : Viability and vigour

127, The two most important factors influencing the life span of seeds under storage are
Ans : Relative humidity andtemperature

128. Seed certification involves
Ans : 5 phase

129. Tetrazolium test determine the level of activities of enzyme
Ans : Dehydrogenase

130. ISTA was established in the year
Ans : 1924

131, Seedlessness in fruits is called as
Ans : Parthenocarpy

132. Naked seeds are generally found in
Ans : Gymnosperm

133. NBPGR is located at
Ans : New Delhi

134. IRRI is located at
Ans : Philippines

135. An organism which has acquired a heritable variation as a result of sudden change in the hereditary material is called as
Ans : Mutation

136. A polyploidy having chromosome sets from different sources such as different species is known as
Ans : Allopolypoid

137. The increased performance ofF1 hybrid over its parents is called
Ans : Heterosis

138. Ratio of the genetic variance to the total variance which determines the relationship between phenotype and genotype is known as
Ans : Heritability

139. Mass seLection method of breeding is particularly, useful for improvement of
Ans : Self pollinated trees

140. Emasculation of flower is necessary for hybridization in
Ans : Self pollinated plants

141. Mendal did not face the problem of linkage, however, seven characters studied by him were located on chromosomes
Ans: Four

142. Test cross ratio in case of duplicate gene action will be
Ans : 3:1

143. A: T ration in one stand of DNA duplex is 2, what will be the ratio in the complementary strand?
Ans : 0.5

144. The term gene coined by
Ans : Johannsen

145. Multiple effects of a single gene is known as
Ans : Pleiotropy

146. An example of a heterozygous but homogenous population is
Ans : Hybrid variety

147. The cross of an inbred line with an open pollinated variety is known as
Ans : Top cross

148. The first artificial hybrid was made b
Ans : ThomasFairchild

149. Double cross plan was suggested by
Ans : Jones

150. Who gave the concept of circular chromosome in *E.coli?*
Ans : Cairns

151. Sex in plants was discovered by
Ans : Camerarious

152. Concept of epigenesist was proposed by
Ans : Wolf

153. Origin of species was published in
Ans: 1859

154. Germplasm theory was proposed by
Ans : Weisman

155. Cell linkage theory was proposed by
Ans : Virchow

156. The chromosomes are most condensed at
Ans : Anaphase

157. DNA synthesis is takes place in stage
Ans : S phase

158. In the DNA double helix G/A is equal to
Ans:C/T

159. The number of bases in the single turn of the DNA helix is
Ans : 10

160. The common form of DNA present in the living organisms is the
Ans: B Form

161. DNA polymerase I was discovered by
Ans : Kornberg

162. The DNA replicating enzyme in bacteria is
Ans: DNA Pol III

163. Protein involved in keeping the DNA single stranded during replication is
Ans : SSB protein

164. Okazaki fragments are formed due to synthesis of DNA in direction
Ans: 5’ — 3’

165. Determination of adenine leads to the formation of
Ans : Hypoxanthine

166. The resulting two daughter cells produced by bridge — breaking fusion cycle will have
Ans : Duplication and deficiency

167. Nucleolus plays important role in production of
Ans: rRNA

168. An amino acid which has related to drought tolerant in plants is?
Ans : Proline

169. in green plants, photosynthesis takes place in
Ans : Chioroplasts

170. Site of protein synthesis in a cell is
Ans : Ribosomes

171.Which cell organelle is having digestive enzymes?
Ans : Lysosomes

172. Who discovered X-rays first?
Ans : Roentgen

173. The defective copies of normal genes are known as
Ans : Pseudo genes

174. The term recon, muton and cistron were coined in 1957 by
Ans : Benzer

175. The variance which is fixable is
Ans : Additive

176. During mitosis, chromosomal DNA duplication takes place in
Ans : Interphase

177. The cell walls of fungi are made up of
Ans : Chitin

178. Antibiotic that cause mis — incorporation of amino acids in protein synthesis is called
Ans Puromycin

179. DNA as a genetic element was shown by the experiment of
Ans Avery, Macleod and McCarty

180. The inherent capacity of an organism to increase in numbers under ideal condition is known as
Ans Biotic potential

181. rDNA analysis provides information on
Ans Phylogenetic relationships

182. cryopreservation is not suitable for preserving organisms which are
Ans : Aerobic

183. Two species belonging to the same genus are identical in morphology which of the structural characteristics is more useful in distinguishing such species they are
Ans Chromosome number and morphology

184. The most important factor which determines stability of seed is
Ans Moisture

185. In seed testing, the grow out test is conducted to determine the
Ans : Genetic purity

186, Translocation of carbohydrates mostly takes place in the form of
Ans : Sucrose

187. IPGRI headquarter is situated in
Ans : Rome, Italy

188. A fertilized mature ovule which possesses an embryonic plant, food reserve and productive coating in a plant species is
Ans: Seed

189. The visible scar (spot) observed on a seed at the place of the detachment from seed stalk is known as
Ans : Hhllum

190. Golgi apparatus is also known as
Ans : Dictyosome

191. Law of homologus series is given by
Ans : Vavilov

192. Transposable elements were discovered by
Ans : McClintock

193. Wobble hypothesis was given by
Ans : Crick

194, Gene for gene hypothesis is proposed by
Ans : Flor

195. Pure line variety is a
Ans : Homozygous — Homogenous

196. Land race are
Ans : Local varieties

197. Lysine is limiting amino acids are
Ans Cereals

198. Triploids are normally
Ans : Sterile

199. Cells without cell wall are known as
Ans : Protoplasts

200. Inability of a hybrid to produce viable offspring is known as
Ans : Hybrid sterility

201. Term heterosis was introduced by
Ans : Shull (1914)

202. Highest uniformity is present in
Ans : Single cross

203. The methods used for fixation of heterosis are
Ans : Apomixes, Polyploid and Asexual reproduction

204. In cross pollinated species, a true breeding line obtained by continuous inbreeding is ‘;alled as
Ans : Inbred

205. Progeny of a single plant obtained by asexual reproduction is
Ans Clone

206. The progeny of a single homozygous self pollinated plant is known as
Ans : Pure line

207. The pure line theory was proposed by
Ans Johannsen

208. The term genetics is coined by
Ans : Bateson

209. The phenomenon of linkage was first observed by
Ans : Bateson and Punnet

210. The ‘one gene one enzyme’ hypothesis was proposed by
Ans : Beadle and Tatum

211. The immediate effect of pollen on the character of endosperm is called as

Ans : Xenia

212. The individuals with 2n ± I chromosomes are commonly known as
Ans : Aneuploid

213. The equilibrium in random mating population is disturbed by selection, mutation, inbreeding, migration and genetic drift in which situation the population may regain its original composition is called
Ans : Mutation

214. The smallest unit of genetic material that can not be divided but is interchangeable among units refers to
Ans : Recon

215. In back cross breeding repeated crossing with recurrent parent is done to
Ans : Completely recover the genotype of recurrent parent

216. In apomixes, the progeny closely resembles
Ans : Female parent

217. Superiority of F1 hybrid over a standard commercial variety is called as
Ans : Economic heterosis

218. Mutation due to change in gene because of
Ans Loss,Degeneration and Addition

219. The chemical mutagens usually induce
Ans : Gene mutation

220. A cybrid is a
Ans : Hybrid formed by cell division

221. Reversion occurs due to
Ans : Mutation

222. Physical basis of life is
Ans : Protoplasm

223. The capacity of a cell to develop into a complete plant is called
Ans : Totipotency

224. Colchicine interfers with
Ans : Formation of spindle fibres

225. Coichicines is generally used for
Ans : Chromosome doubling

226. Sum total of deleterious genes present in species is known as
Ans : Genetic load

227. Chromosomal theory of inheritance was given by
Ans Sutton and Boveri

228. Reverse transcription is also known as
Ans : Teminism

229. RFLP is
Ans : Molecular marker

230. Double helical structure of DNA was shown by
Ans : Watson and crick

231. The F2 ratio in complementary gene action is
Ans: 9:7

232. Who is known as father of genetics?
Ans : Mendel

233. The test cross ratio in complementary gene action is
Ans : 1:3

234. The F2 ratio in polymeric gene action is
Ans : 9:6:1

235. Translocation involves
Ans : Homologous chromosomes

236. chromosomes other than sex chromosomes are called
Ans : Autosomes

237. An exchange of genetic material between homologous chromosomes is called as
Ans : Crossing over

238. The word mutation was first used by
Ans : Morgan

239. Coupling and repulsion hypothesis was given by
Ans : Bateson

240. Theory of Gemmules was proposed by
Ans : Darwin

241. Incompatibility is very common in plant kingdom was reported by
Ans : Lewis

242. Improved seed distributed to farmers for commercial cultivation is
Ans : Certified

243. The tissue which nourish microspores is
Ans : Tapatum

244. The molecular markers which are codominant in nature are
Ans : SSR and RFLP

245. A virus free clone from a virus infected plant can be obtained by
Ans : Meristem culture

246. In mitosis, longest phase is
Ans : Prophse

247. Rediscovery of mendelian principles in the year
Ans : 1900

248. Term genetics was given by
Ans : Betson (1905)

249. The lines are homozygous and homogenous in nature called as
Ans : Pure line

250. Two lines different for a single locus called
Ans : iso genic line

251. Laws of heredity were first discovered by
Ans Mandel

252. The term Germplasm was first used by
Ans: Weismann (1834-1914)

253. NBPGR established in
Ans: 1976

254. The term genetic resources was coined by
Ans : Frankel

255. The term parthogenesis was coined by
Ans : Owen

256. Development of seed by self pollination refers to
Ans : Autogamy

257. Chromosome was discovered by
Ans : Strasburger

258. Longest phase of Mitosis is
Ans : Prophase

259. Microtubles were discovered by
Ans : De Robertis and Franchi in 1953

260. Concept of pure line theory was developed by
Ans : Johansen

261. Largest organelle in Eukaryotic cell
Ans : Nucleus

262. Earth summit held at
Ans : Rio de Janeiro

263. Quantification of genetic diversity parameters and genome analysis of forest trees can be made through technique.
Ans : Molecular marker

264. Example for endangered tree genetic resource is
Ans : Santalum *album, Pterocarpus Santalinus*

265. Sanctuaries for Rhododendron and orchids have been established in state
Ans : Sikkim

266. CITES stands for
Ans: Convention on International Trade in Endangered Species of Wild Fauna and Flora

267. The migration of alleles from one population or species br movement of pollen or seed into another is known as
Ans: Gene flow

268. In India, large forest area is covered by forest type
Ans : Dry deciduous forest

269. PPV&FRA refers to
Ans : Protection of Plant Varieties and Farmers' Right Act

270. The Ramsar Convention relates to genetic resources
Ans : Wetland

271. First Biosphere Reserve in India is
Ans : Gulf of Mannar Marine Biosphere Reserve

272. GIS refers to
Ans : Geographic Information System

273. Quantification of genetic diversity parameters and genome analysis of forest trees can be made through technique.
Ans : Molecular marker

274: Uiólöglcal diversity act was enacted during the year
Ans : 2002

275. Sanctuaries for Rhododendron and orchids have been established in state
Ans: Sikkim

276. IUCN stands for
Ans : International Union for Conservation of Nature and Natural Resources

277. The migration of alleles from one population or species br movement of pollen or seed into another is known as
Ans : Gene flow

278. Forest types of India (1968) was prepared by
Ans : Champion and Seth

279. TRIPS refers to
Ans : Trade related intellectual property rights

280 numbers of flora are found in India.
Ans : 45000

281. Earth summit held at Rio de Janeiro, Brazil during the year
Ans : 1992

282, India is one of the of biological diversity in the world
Ans : 12 centres

283. Variation is also essential for selection and to meet present and future human needs.
Ans Breeding

284. The term used for area rich in endemic plants and containing high diversity of species.
Ans : Biodiversity hotspots

285. GATT established in the year of
Ans : 1984

286. DFSC stands for
Ans : Danida Forest Seed Centre

287. In bamboo genetic resource is second rich country in the world
Ans : India

288. The Western Ghats spread to number of states in India.
Ans : 6

289. The genetic diversity refers to
Ans : Variation of genes within species

290, Head quarters of WTO situated at
Ans : Geneva

291. The primitive cultivars which were selected and cultivated by farmers for many generations is called
Ans Land races

292. Gradual reduction in genetic variability due to elimination of various genotypes is known as
Ans : Genetic erosion

293. The term used for trips arranged for collection of germplasm from different areas
Ans : Plant exploration

294. TRIPS denotes
Ans : Trade-Related Aspects of Intellectual Property Rights

295. Gene flow or gene transfer takes place on the species level through process called
Ans : Introgression

296. The germplasm which is collected or received from other countries is called as
Ans : Exotic collection

297. India ratified the Kyoto protocol in year
Ans : 2002

298. The prophylactic measure that is used to prevent the entry of new diseases, insects and

weeds from other countries is called as
Ans : Quarantine

299. PPV&FRA refers to
Ans : Protection of Plant Varieties and Farmers' Right Act

300. The variety of species within a region is referred as
Ans : Species diversity